SKYWORTH
Your Best Friend of Energy-Saving Solutions

# SWH
# 5/6/8/10/12/15/20kW
## Three Phase Hybrid Inverter

**MODEL**

SWH5000L-T1-G2 | SWH6000L-T1-G2 | SWH8000L-T1-G2 | SWH010KL-T1-G2

SWH012KL-T1-G2 | SWH015KL-T1-G2 | SWH020KL-T1-G2




**High Performance**
· 20A per string & 200% PV oversizing
· Dual independent battery inputs with 416A

**Smart Control & Monitoring**
· Seamless switching < 4ms
· Generator compatibility and charge for battery

**Superb Safety & Reliability**
· AI-driven AFCI optional[1]
· Type II SPD on both DC&AC sides
· IP66 ingress protection

**Flexible & Adaptable Applications**
· Unbalanced output up to 150%
· Compatible with both Li-Ion/Lead-acid batteries

1: Optional functions or devices are purchased separately

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| Technical Data | SWH5000L-T1-G2 | SWH6000L-T1-G2 | SWH8000L-T1-G2 | SWH010KL-T1-G2 | SWH012KL-T1-G2 | SWH015KL-T1-G2 | SWH020KL-T1-G2 |
|---|---|---|---|---|---|---|---|
| **Battery Side** | | | | | | | |
| Battery Type | | | | LFP ($LiFePO_4$) / Lead-acid | | | |
| Nominal Battery Voltage (V) | | | | 48 | | | |
| Battery voltage range (V) | | | | 40 ~ 60 | | | |
| Start-up Voltage (V) | | | | 30 | | | |
| Number of Battery Inputs | 1 | 1 | 1 | 1 | 1 | 2 | 2 |
| Max. Continuous Charging Current (A) | 125 | 135 | 175 | 220 | 250 | 165 | 208 / 208 |
| Max. Continuous Discharging Current (A) | 125 | 135 | 175 | 220 | 250 | 165 | 208 / 208 |
| Max. Charging Power (W) | 5000 | 6000 | 8000 | 10000 | 12000 | 15000 | 20000 |
| Max. Discharging Power (W) | 5500 | 6600 | 8800 | 11000 | 13200 | 16500 | 22000 |
| **PV Side** | | | | | | | |
| Max. Input Power (kW) | 10 | 12 | 16 | 20 | 24 | 30 | 40 |
| Max. Input Voltage (V)*1 | | | | 1000 | | | |
| MPPT Operating Voltage Range (V)*3 | | | | 150 ~ 850 | | | |
| Start-up Voltage (V) | | | | 180 | | | |
| Nominal Input Voltage (V) | | | | 620 | | | |
| Max. MPPT Current (A) | 20 / 20 | 20 / 20 | 20 / 20 | 20 / 20 / 20 | 20 / 20 / 20 | 20 / 20 / 20 / 20 | 20 / 20 / 20 / 20 |
| Max. MPPT Short Circuit Current (A) | 26 / 26 | 26 / 26 | 26 / 26 | 26 / 26 / 26 | 26 / 26 / 26 | 26 / 26 / 26 / 26 | 26 / 26 / 26 / 26 |
| Number of MPPTs | 2 | 2 | 2 | 3 | 3 | 4 | 4 |
| Number of Strings per MPPT | 1 / 1 | 1 / 1 | 1 / 1 | 1 / 1 / 1 | 1 / 1 / 1 | 1 / 1 / 1 / 1 | 1 / 1 / 1 / 1 |
| **AC Side (On-grid)** | | | | | | | |
| Nominal Power (W) | 5000 | 6000 | 8000 | 10000 | 12000 | 15000 | 20000 |
| Max. Power (W) | 5500 | 6600 | 8800 | 11000 | 13200 | 16500 | 22000 |
| Nominal Apparent Power Output to Grid (VA) | 5000 | 6000 | 8000 | 10000 | 12000 | 15000 | 20000 |
| Max. Apparent Power to Utility Grid (VA) | 5500 | 6600 | 8800 | 11000 | 13200*4 | 16500*4 | 22000*4 |
| Nominal Apparent Power from Grid (VA) | 5000 | 6000 | 8000 | 10000 | 12000 | 15000 | 20000 |
| Max. Apparent Power from Grid (VA) | 48300 | 48300 | 48300 | 48300 | 48300 | 48300 | 48300 |
| Nominal Voltage (V) | | | | 400 / 380, 3L / N / PE | | | |
| Voltage Range (V) | | | | 170 ~ 290 | | | |
| Nominal Frequency (Hz) | | | | 50 / 60 | | | |
| Frequency Range (Hz) | | | | 45 ~ 65 | | | |
| Max. Current to Grid (A) | 7.9 / 8.3 | 9.6 / 10 | 12.8 / 13.4 | 15.9 / 16.7 | 19.1*5 / 20 | 23.9*5 / 25 | 31.9 / 33.3 |
| Max. Current From Grid (A) | | | | 70 | | | |
| Nominal Current From Grid (A) | 7.5 / 7.2 | 9.1 / 8.7 | 12.1 / 11.6 | 15.2 / 14.5 | 18.2 / 17.4 | 22.7 / 21.7 | 30.3 / 29 |
| Nominal Current to Grid (A) | 7.5 / 7.2 | 9.1 / 8.7 | 12.1 / 11.6 | 15.2 / 14.5 | 18.2 / 17.4 | 22.7 / 21.7 | 30.3 / 29.0 |
| Power Factor | | | | ~1 (Adjustable from0.8 leading~0.8 lagging) | | | |
| THDI | | | | <3% | | | |
| **Back-up Side** | | | | | | | |
| Nominal Output Apparent Power (VA) | 5000 | 6000 | 8000 | 10000 | 12000 | 15000 | 20000 |
| Max. Output Apparent Power (VA) | 5500 (10000, 10s) | 6600 (12.0, 10s) | 8800(16000, 10s ) | 11000 (20000, 10s ) | 13200 (24000, 10s ) | 16500 (30000, 10s ) | 22000 (40000, 10s ) |
| Max. Output Apparent Power with Grid (kVA) | | | | 48.3 | | | |
| Max. Output Current (Bypass) | | | | 70 | | | |
| Nominal Output Voltage (V) | | | | 400 / 380, 3L / N / PE | | | |
| Nominal Output Frequency (Hz) | | | | 50 / 60 | | | |
| **Efficiency** | | | | | | | |
| Max. Efficiency | 97.8% | 97.8% | 97.8% | 97.8% | 97.8% | 97.9% | 97.8% |
| European Efficiency | 97.0% | 97.0% | 97.1% | 97.3% | 97.3% | 97.3% | 97.4% |
| Max. Battery to AC Efficiency | | | | 95.5% | | | |
| MPPT Efficiency | | | | 99.9% | | | |
| **Protection** | | | | | | | |
| Battery Reverse Polarity Protection | ○ | ○ | ○ | ○ | ○*6 | ○*6 | ○*6 |
| DC Switch | | | | ● | | | |
| DC Surge Protection | | | | Type II | | | |
| AC Surge Protection | | | | Type II | | | |
| AFCI | | | | ○ | | | |
| Rapid Shutdown | | | | ○ | | | |
| Remote Shutdown | | | | ● | | | |
| **General Data** | | | | | | | |
| Operating Temperature Range (°C) | | | | -35 ~ +60 | | | |
| Relative Humidity | | | | 0 ~ 95% | | | |
| Max. Operating Altitude (m) | | | | 4000 | | | |
| Cooling Method | | | | Smart Fan Cooling | | | |
| User Interface | | | | LCD | | | |
| Communication with BMS | | | | CAN | | | |
| Communication | | | | WIFI+LAN+Bluetooth, 4G, RS485, CAN | | | |
| Communication Protocols | | | | Modbus RTU (RS485), Modbus TCP / IP (Ethernet), Sunspec Modbus RTU | | | |
| Weight (kg) | 42.2 | 42.2 | 42.2 | 45.3 | 45.3 | 49.7 | 51.2 |
| Dimension (W × H × D mm) | | | | 756 × 551 × 258 | | | |
| Noise Emission (dB) | | | | <45 | | | |
| Power Self-consumption at Night (W) | | | | <15 | | | |
| Ingress Protection Rating | | | | IP66 | | | |
| Anti-corrosion Class | | | | C4 | | | |
| Mounting Method | | | | Wall Mounted | | | |

● Integrated ○ Optional

*1: The Li-Ion battery usually contain two mainstream type: LFP and Ternary Lithium battery.
*2: When the input voltage is greater than 980V, the inverter will enter standby mode, and when the voltage returns to below 950V the inverter will return to normal operation.
*3: Please refer to the user manual for the MPPT Voltage Range at Nominal Power.
*4: For Brazil and Chile, the Max. Power is the same with the Nominal Power.
*5: For Brazil, for SWH012KL-T1-G2, SWH015KL-T1-G2 and SWH020KL-T1-G2, the Max. Current to Grid is 18.2A@220V, 22.7A@220V and 30.3A@220V respectively.
*6: For Brazil, for SWH012KL-T1-G2, SWH015KL-T1-G2 and SWH020KL-T1-G2, the Battery Reverse Polarity Protection is integrated.
*7: AFCI is integrated in Brazil.
*: Please visit SKYWORTH website for the latest certificates.
*: All pictures shown are for reference only. Actual appearance may vary.

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